

L-16: Probability Notes

Recall: Hölder's Inequality

Let X and Y be two random variables. Then

$$\mathbb{E}[|XY|] \leq \|X\|_p \cdot \|Y\|_q$$

where $p > 1$ and p, q are the Hölder conjugate such that

$$\frac{1}{p} + \frac{1}{q} = 1$$

(This implies $q = \frac{p}{p-1}$).

Proof

Let $A = \|X\|_p$ and $B = \|Y\|_q$. We assume $0 < A, B < \infty$.

$$\text{T.S.} \quad \|X \cdot Y\|_1 \leq A \cdot B \iff \left\| \frac{X}{A} \cdot \frac{Y}{B} \right\|_1 = 1$$

Fact: Little Hölder's Inequality / Young's Inequality

Let $a, b > 0$ and $p > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$a \cdot b \leq \frac{a^p}{p} + \frac{b^q}{q}$$

Continuing the proof of Hölder's

Apply Young's inequality pointwise for each ω : Let $a = \frac{|X(\omega)|}{A}$ and $b = \frac{|Y(\omega)|}{B}$.

$$\frac{|X(\omega) \cdot Y(\omega)|}{AB} \leq \frac{1}{p} \frac{|X(\omega)|^p}{A^p} + \frac{1}{q} \frac{|Y(\omega)|^q}{B^q}$$

Take expectation on both sides:

$$\frac{1}{AB} \mathbb{E}[|XY|] \leq \frac{1}{p} \frac{\mathbb{E}[|X|^p]}{A^p} + \frac{1}{q} \frac{\mathbb{E}[|Y|^q]}{B^q}$$

Recall by definition, $A^p = \mathbb{E}[|X|^p] = \|X\|_p^p$ and $B^q = \mathbb{E}[|Y|^q] = \|Y\|_q^q$.

$$\frac{1}{AB} \mathbb{E}[|XY|] \leq \frac{1}{p} \frac{A^p}{A^p} + \frac{1}{q} \frac{B^q}{B^q}$$

$$\frac{1}{AB} \mathbb{E}[|XY|] \leq \frac{1}{p} + \frac{1}{q} = 1$$

$$\mathbb{E}[|XY|] \leq A \cdot B = \|X\|_p \cdot \|Y\|_q$$

Proof of Young's Inequality: $f(t) = e^{\lambda t}$ is convex, $\lambda > 0$.

$\frac{1}{p} + \frac{1}{q} = 1 \Rightarrow$ Notice $\left(\frac{1}{p}, \frac{1}{q}\right)$ is a pair of convex combination coefficients.

$$\underbrace{e^{\frac{s}{p}}}_a \quad \text{and} \quad \underbrace{e^{\frac{t}{q}}}_b$$

$$e^{\frac{s}{p} + \frac{t}{q}} \leq \frac{1}{p}e^s + \frac{1}{q}e^t \quad (\text{convex for any } s, t).$$

$$s = b \log a, \quad t = a \log b$$

L_p Spaces

The L_p space is defined as:

$$L_p = \{X : \Omega \rightarrow \mathbb{R} \mid \mathbb{E}[|X|^p] < \infty\}$$

It has the following properties:

- It is a vector space.
- The L_p -norm is a function $\|\cdot\|_p : L_p \rightarrow [0, \infty)$ defined by

$$\|X\|_p = (\mathbb{E}[|X|^p])^{1/p}$$

- $\|X\|_p \geq 0$.
- $\|c \cdot X\|_p = |c| \cdot \|X\|_p, \quad \forall X \in L_p, \forall c \in \mathbb{R}$.
- **(Minkowski's Inequality)** $\|X + Y\|_p \leq \|X\|_p + \|Y\|_p, \quad \forall X, Y \in L_p$.
- $\|X\|_p = 0 \iff X = 0$ almost surely.

The L_2 Space

The L_2 space is a special case:

$$L_2 = \{X : \Omega \rightarrow \mathbb{R} \mid \mathbb{E}[X^2] < \infty\}$$

The norm is $\|X\|_2 = (\mathbb{E}[X^2])^{1/2} = \sqrt{\mathbb{E}[X^2]}$.

Note: IF X is s.t. X in L_2 and $\mathbb{E}[X]=0$ then, $\|X\|_2 = \text{SE}[X]$ On L_2 , an inner product is defined:

$$\begin{aligned} \langle \cdot, \cdot \rangle : L_2 \times L_2 &\rightarrow \mathbb{R} \\ (X, Y) &\mapsto \langle X, Y \rangle = \mathbb{E}[XY] \end{aligned}$$

Minimization Problems

1. L_2 Minimization

If $X \in L_2$, what is the constant c that minimizes $\mathbb{E}[(X - c)^2]$?

- **Answer:** $c = \mu = \mathbb{E}[X]$

2. L_1 Minimization

If $X \in L_1$, what is the constant c that minimizes $\mathbb{E}[|X - c|]$?

- **Answer:** $c = m = \text{median of } X$

Summary

$$\sigma^2 = \text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \min_{c \in \mathbb{R}} \mathbb{E}[(X - c)^2]$$
$$\mathbb{E}[|X - m|] = \min_{c \in \mathbb{R}} \mathbb{E}[|X - c|]$$

Example: $\mathbb{E}[|X - m|] \leq \sigma$

Convergence

Question: Suppose $X_n \rightarrow X$ almost surely. Is it always true that $\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]$?

- **Answer:** NO.

Note: We know that it is not necessary to have $\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]$ even if $X_n \rightarrow X$ a.s.

Counterexample

Let $U \sim \text{Uniform}(0, 1)$. Define the sequence of random variables $X_n = n \cdot \mathbb{I}(U \leq 1/n)$.

$$X_n(\omega) = \begin{cases} n & \text{if } U(\omega) \leq 1/n \\ 0 & \text{otherwise} \end{cases}$$

- **Almost Sure Convergence:** For any ω , $U(\omega) = u \in (0, 1)$. For $n > 1/u$, we have $1/n < u$, so $X_n(\omega) = 0$. Therefore, $\lim_{n \rightarrow \infty} X_n(\omega) = 0$ for all ω . So $X_n \rightarrow 0$ almost surely.
- **Expectation:** $\mathbb{E}[X_n] = n \cdot P(U \leq 1/n) + 0 \cdot P(U > 1/n) = n \cdot (1/n) = 1$.

So, we have $X_n \geq 0$, $X_n \rightarrow 0$ a.s., but $\mathbb{E}[X_n] = 1$ for all n , so $\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = 1 \neq 0$.

Monotone Convergence Theorem (MCT)

A key condition for swapping limit and expectation is non-negativity and monotonicity.

If $X_n \geq 0$ and $X_n \uparrow X$ (monotonically increasing w by w) almost surely, then $\mathbb{E}[X_n] \uparrow \mathbb{E}[X]$.